

Java

Access specifier & file handling

```
package package1;

public class user {

    // access specifier

    // private => accessible within the same class
    // default => accessible within the same package
    // protected => accessible within the same package + subclasses (via inheritance)
    // public => access anywhere in the code


    //          same class      same package      different package
    // private      yes          No                No
    // default      yes          Yes               No
    // protected    yes          yes              Yes(via inheritance)
    // public       Yes          yes              yes


    private String name = "Padma";

    int rollno = 123;

    protected int salary = 50000;

    public String role = "Software developer";

    private void display(){
        System.out.println("hello there");
    }
    public void setName(String name){
        this.name = name;
        display();
    }

    public String getName(){
        return name;
    }
}
```

```
package package1;

public class package1Example {

    public static void main(String[] args) {

        user obj = new user();

        obj.setName("FIIT");
        System.out.println(obj.getName());
        System.out.println(obj.rollno);
        System.out.println(obj.salary);

        System.out.println(obj.role);
    }
}
```

```
package package2;

import package1.user;

public class admin extends user{

    public static void main(String[] args) {
        user obj1 = new user();

        // System.out.println(obj1.rollno);
        // System.out.println(obj1.salary);

        System.out.println(obj1.role);

        admin obj = new admin();

        System.out.println(obj.salary);
    }
}
```

```
import java.io.File;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class FileHandling {
    public static void main(String[] args) throws IOException {

        File fs = new File("src/example.txt");
        File fs2= new File("src/files/files1/files2/files3/pdfFile.pdf");
        fs2.createNewFile();

        File fs3= new File("src/files/files1/files2/docFile.docx");
        fs3.createNewFile();

        File fs4= new File("src/files/files1/pptFile.pptx");
        fs4.createNewFile();

        fs.createNewFile();

        if(fs.exists()){
            System.out.println("File is created successfully");

            System.out.println(fs.getName());

            System.out.println(fs.canWrite());

            System.out.println(fs.canRead());

            System.out.println(fs.getAbsolutePath());
            System.out.println(fs.length());
            // fs.delete();
        }else{
            System.out.println("file is not found");
        }

        File fs1 = new File("src/files/files1/files2/files3");

        // fs1.mkdir();

        fs1.mkdirs();

        try {
            FileWriter write = new FileWriter("src/example.txt");

            write.write("THis is demo file ");
        }
    }
}
```

```
        write.close();

    } catch (Exception e) {
        // TODO: handle exception
        System.out.println(e.getMessage());
    }

    // read a file

    try {
        Scanner sc = new Scanner(fs);

        while (sc.hasNextLine()) {
            String data = sc.nextLine();
            System.out.println(data);
        }

    } catch (Exception e) {
        // TODO: handle exception
    }

    // append
    try {

        FileWriter fileAppend = new FileWriter("src/example.txt", true);

        fileAppend.write("\n Appended file");

        fileAppend.close();

    } catch (Exception e) {
        // TODO: handle exception
    }

}

}
```

Theory

1. Access Specifiers in Java

Definition: Access specifiers define the visibility of classes, methods, and variables.

Types:

- **private** → accessible only within the same class.
- **default (no modifier)** → accessible within the same package.
- **protected** → accessible within the same package + subclasses (via inheritance).
- **public** → accessible everywhere.

Comparison Table:

Specifier	Same Class	Same Package	Different Package	Subclass (different package)
private	✓	✗	✗	✗
default	✓	✓	✗	✗
protected	✓	✓	✗	✓
public	✓	✓	✓	✓

2. File Handling in Java

Definition: File handling allows reading, writing, and managing files using the java.io package.

Key Classes:

- **File** → represents file/directory path.
- **FileWriter** → writes data to file.
- **Scanner** → reads data from file. **Key Methods:**
 - **createNewFile()** → creates a new file.
 - **exists()** → checks if file exists.
 - **delete()** → deletes file.
 - **write()** → writes content.
 - **append** → adds content without overwriting.

Coding Round Questions

- Access specifiers → Explain how private, default, protected, and public work with examples.
- Inheritance with protected → Show how protected members are accessible in subclasses across packages.
- File creation → Write a program to create a new file and check its properties.
- File writing → Write text into a file using FileWriter.
- File reading → Read contents of a file using Scanner.
- File append → Append text to an existing file.

Machine Coding Round Challenge

Problem: Employee Record System

Requirements:

- Create a Employee class with fields: id, name, salary.
- Use **access specifiers**:
 - private for id.
 - protected for salary.
 - public for name.
- Create a file employee.txt to store employee details.
- Write methods to:
 - Add employee details to file.
 - Read employee details from file.
 - Append new employee records.
- Demonstrate inheritance by creating a subclass Manager that can access salary.